MARK SCICLUNA

Senior Software Engineer - Mobile, Spatial & Applied AI

Melbourne, Australia | dicarkus@hotmail.com | +61 408 513 913 | GitHub | LinkedIn | mscicluna.com

SUMMARY

Senior software engineer with 10+ years of experience across mobile, spatial computing, web and applied AI. Experienced in taking products from early concept and architecture through implementation, release and ongoing support, with hands-on work across Swift/SwiftUI, React, Flutter, Kotlin, Python and .NET. Recent projects include spatial applications for Apple Vision Pro and Samsung Galaxy XR, alongside deployed AI systems using multi-agent orchestration, MCP, multi-provider LLMs and structured AI workflows.

CORE SKILLS

Mobile & Spatial: Swift, SwiftUI, Flutter, Kotlin, visionOS, RealityKit, ARKit, Android XR

Applied AI: Python, Multi-agent orchestration, prompt/system design, NLP, dynamic agent generation, tool use via MCP, structured output validation, retry/fallback logic, RAG, model evaluation, observability, local/private inference, AI-assisted workflow design

Web & Backend: React, TypeScript, Vue, Node.js, ASP.NET Core, PostgreSQL, REST

Delivery: System architecture, state management, CI/CD, Jenkins, Docker, AWS, App Store / Google Play release

EXPERIENCE

Senior Software Engineer - Rhizomenet Pty Ltd

2017–2026 | ZOME cross-platform geospatial platform

Worked across Flutter, iOS, Android, web and spatial computing, with responsibility for client architecture, UI systems, state management, media behaviour, performance, release engineering and platform adaptation. Regularly helped turn evolving product ideas into practical implementations.

- Built the ZOME visionOS client from scratch, owning application architecture, state, immersive spaces, media, interaction systems and release engineering.
- Developed an arm-anchored RealityKit interface using skeletal hand tracking, quaternion transforms and independent coordinate spaces; added smoothing, bounds handling and occlusion fixes for stable gesture-driven interaction.
- Adapted Hussein Chalayan's spatial UI concept to practical visionOS constraints and shipped for the Apple Vision Pro launch; also delivered ZOME for Samsung Galaxy XR.
- Designed a virtualised/indexed Flutter list approach and refactored Redux/global state to reduce rendering pressure, memory use, unnecessary rebuilds and UI coupling.
- Maintained Jenkins pipelines, macOS build agents, Xcode upgrades, provisioning, signing and deployment automation.

Senior Developer / Applied AI Developer - Smooth Software (carkus.com)

2025–Present | Deployed AI, web and office administration systems

Design and build deployed software across AI, web and document-generation systems, covering product architecture, backend/API development, agentic workflows, structured output handling, provider orchestration and deployment.

- Built Agent One, a deployed multi-agent system that decomposes requests, creates specialist worker agents at runtime and assigns tools dynamically through a vetted MCP registry.
- Implemented local-first inference, provider fallback, retries, structured-output validation, response evaluation and observability across token usage, latency and provider failures.
- Built MarkdownForms end to end: a custom Markdown form DSL, visual builder, AI-assisted generation/editing and fillable PDF output, with safeguards around destructive AI edits.
- Built JobFit, a modular .NET job-fit system using structured LLM appraisal, deterministic scoring, provider fallback, schema validation and parallel recommendation analysis.

Software Developer - RMIT

2007–2014 | E-learning platforms for ESL and aviation

Designed, built and supported e-learning software for ESL and aviation training, covering interface design, reusable activity systems, XML-driven course data, backend integration, classroom deployment and ongoing support.

- Owned eCourse end to end: system architecture, XML data model, interaction/UI design, implementation, deployment and ongoing support.
- Built approximately four e-learning platforms; eCourse supported roughly 400 activities through a reusable XML-driven template system.
- Enabled instructional designers to create and update course content without hand-coding each screen, while template improvements propagated across all matching activities.

SELECTED PROJECTS

ZOME - Cross-platform geospatial application spanning Flutter, iOS, Android, web, Apple Vision Pro and Samsung Galaxy XR, including custom spatial UI, state-management refactoring, performance optimisation and release engineering.
Agent One - Dynamic multi-agent orchestration, MCP tooling, local/private inference, provider fallback and evaluation.
MarkdownForms - AI-assisted form generation using a custom Markdown DSL, visual builder and fillable PDF pipeline.
JobFit - Modular AI job-fit evaluation with deterministic scoring, structured validation and multi-provider orchestration.

EDUCATION

MSc Computer Science - Distinction, RMIT University

Thesis: Search Efficiency in Robotic Task Planning - heuristic search, AI task re-planning, plan repair and PDDL-based planning systems.